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A Note on the Estimation of the Energy Expenditure of Sporting Cyclists

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Estimates are made of the energy expenditure necessary for propelling mounted cyclists at speeds up to the maximum recorded for prolonged periods of riding, unpaced, on level roads. Two methods of calculation are presented, one based on the oxygen consumption of cyclists actually riding on the road and recorded in the literature, the other upon recorded data concerning the tractive resistances exhibited by moving mounted cyclists.

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1. Introduction

The gross caloric expenditure of riders of bicycles has been investigated by several physiologists: Zuntz (1899), Dill (1954), Edholm (1955), Malhotra *et al.* (1962), and Adams (1967). The technique used has been to measure by means of various portable types of apparatus the actual oxygen consumption of the riding cyclists. The recorded results of this type of work show that a limited range only of riding speeds was used in the experiments: the maximum speed was only 13 miles per hour. In addition, little information is given concerning important aspects of the machines used and the riders' weight and frontal areas presented to the wind. It does not appear, therefore, wise to extrapolate such data in order to estimate the energy expenditure of racing cyclists who adopt crouched riding positions and ride highly efficiently on easy-running bicycles.

It is possible to measure the energy expenditure of riding cyclists by another method which depends upon the determination of the tractive resistance offered by towed cyclists. This dynamometer technique is a very old established experimental method dating back some two hundred years, when the power to move ploughs was a subject of great interest, and resulted in the development of the railway dynamometer car. Towing experiments were carried out a long time ago on cyclists. Sharp (1896) describes some early work, and Chandler (1910) describes later experiments. Nonweiler (1956) measured the wind resistance of mounted racing cyclists suspended in a wind tunnel and derived similar data concerning the tractive resistance offered by cyclists. Davison (1944) discusses methods of determining frontal areas.

This latter type of experimental work described above, has been carried out into the high speed range of 20 and over miles per hour and is on this account more amenable to reliable extrapolation, particularly in view of the fact that the specifications of bicycle and riders characteristics are also quoted.

2. Tractive Power Necessary to Propel Mounted Cyclists

For level running in still air the major resistances opposing the motion of a cyclist are that due to his self-created wind opposition and that due to energy losses in tyre flexion when his wheels roll along the road. This latter loss is termed ' rolling resistance '. The frictional losses in the wheel bearings are very small indeed and negligible for the purposes on hand.

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2.1. *Air Resistance*

On detailed examination of the information already referred to as recorded in the literature for towing or wind tunnel experiments, it appears that the simplest and most appropriate correlation between the major variables is expressible by the equation

$$\underset{\text{lb force}}{\text{Air resistance}} = 0{\cdot}0023 \times \underset{\text{sq ft}}{\text{Frontal area}} \times \underset{\text{mile/h,}}{\text{speed}^2} \tag{1}$$

assuming that the atmospheric conditions are such that the air density is 0·0765 lb per cu ft. The constant in the equation has a value close to that which would be applicable to 'square ended' vehicles like lorries and railway engines. Saloon cars are associated with constants of much lower value.

2.2. *Rolling Resistance*

Experimental data given by Sharp (1896), Chandler (1910), Judge (1925), Kempe (1969) and Whitt (1967) leads to the conclusion that reliable estimations for tyre rolling resistance can be obtained from the use of the equation

$$\underset{\text{lb force per lb weight}}{\text{Rolling resistance}} = 0{\cdot}005 + \frac{(0{\cdot}15)}{\text{inflation pressure lbf/in}^2}, \tag{2}$$

assuming that lightweight tyres of $1\frac{1}{4}$ or $1\frac{3}{8}$ in cross section are used on wheels of 26 in. diameter and that they are running on smooth hard concrete or macadamised roads. Wheels of smaller diameter have increased rolling resistance in inverse proportion.

2.3. *Power Requirements*

The power output necessary to overcome the sum of the air and rolling resistances can be calculated from the equation

$$\underset{\text{h.p.}}{\text{Power}} = \text{Total resistance}\,\frac{(\text{lbf})}{375} \times \text{speed (mile/h)}. \tag{3}$$

Using the above equations the tractive powers necessary to propel two typical types of cyclist under particular conditions have been calculated. (See Table 1.) A racing cyclist is assumed to weigh with machine, 77 Kg (170 lb) and present a frontal area of about $\frac{1}{3}$ sq metre (3·65 sq ft) and with tyre inflation pressures of 100 lbf/in^2. A touring cyclist is assumed to weigh with (heavier) machine, 83 Kg (187 lb) with a frontal area of about $\frac{1}{2}$ sq metre (5·5 sq ft) and with tyre inflation pressures of 50 lbf/in^2. The riders weight (stripped) is, in both cases, 66 Kg (148 lb).

Table 1. Speed, power, oxygen consumption and metabolic heat relationships

Speed, mile/h		Tractive power h.p.	Oxygen consumption litres/min	Metabolic heat cals/min
Racing	Touring			
27	22·5	0·5	4·8	24
25	21	0·4	3·9	19·5
22	18·5	0·3	3	15
19	16	0·2	2·1	10·5
14·5	12	0·1	1·2	6
10·5	8·3	0·05	0·75	3·75
7·2	6	0·025	0·525	2·65
3	1·8	0·008	0·372	1·9
0	0	0	0·3	1·5

3. Oxygen Consumption Tests on Riding Cyclists

Adams (1967) gives a summary of recorded work in the literature. Figure 1 reproduces the information as points on a plot of speed versus gross caloric expenditure.

On Figure 1 are also plotted the power data of Table 1 converted to oxygen consumptions and then to metabolic heat data. A calorific value of 5 cal per litre of oxygen has been assumed. A 'round' value of 0·1 h.p. needing the consumption of 0·9 litres of oxygen per minute has been assumed. (0·1 h.p. is equivalent to 1·068 cals per minute or 7·6 Kg/m per second.) This implies a net efficiency of 23·6%—a slightly lower figure than is commonly accepted for ergometer work because it was thought advisable to recognize that bicycle transmission losses do occur (see Van Dobeln 1954). Wyndham *et al.* (1966) and others suggest a net efficiency of +25%; a Japanese report (1968) gives gross efficiencies of over 25%.

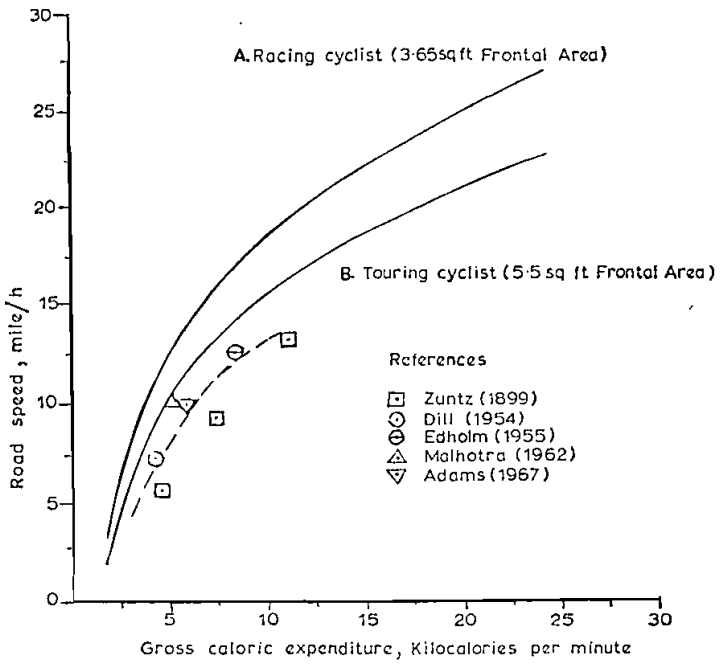

Figure 1. Gross calorific expenditure of various types of cyclist.

4. Summary and Conclusions

4.1. *Metabolic Heat Output Rates*

Examination of Figure 1 shows that there is good agreement between all the data as to the general trend of increase of metabolic heat expenditure with speed of cycling. In addition the displacement of the curves A and B can be expected in that, although scanty, the descriptions given of the machines and riders involved in oxygen consumption tests suggest that relatively inefficient machines and unpractised riders were involved. No doubt relatively soft tyres and upright riding positions were adopted.

It is interesting to note that a commonly encountered speed for utility cycling of 10 miles per hour, as quoted in engineering text books for many

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years as needing the expenditure of about 0·1 h.p., needs an oxygen consumption of rather less than half the maximum for many non-athletic people (2½ litres per minute). This is a phenomenon now accepted widely for the 'safe' working of labourers.

The upper ends of curves A and B appear to give realistic estimates in view of the fact that good class racing cyclists riding unspaced in 'out and home' road time trials achieve 24–25 miles per hour for up to 4 hours on end. Åstrand (1967) states that long distance skiers often work at about 80% of their maximum oxygen consumption rates. The cyclists appear to be using about the same proportionate rates, of probable maximum of 5 litres per minute.

It is probable that record making cyclists who achieve about 26–27 miles per hour do so because they achieve higher oxygen efficiencies for a combination of various reasons. Garry and Wishart (1931) noted that their professional cyclist was a competent pedaller and also tended to use a higher rate of pedalling than the commonly accepted 50 revs/min (said to be comfortable for probably inexperienced subjects). Skilled cyclists are well aware of the need for correct 'gears' (see Brown 1944) and the need for keeping a steady seat. If too great a push is required riders tend to rock about and waste energy. (See Report of the Bicycle Production and Technical Institute, Japan, 1968.) A common gear of 67 in (the equivalent diameter of the driving wheel if it was an ungeared-up wheel), such as a number of utility bicycles are equipped with, involves the rider pedalling at about 50 revs/min at 10 miles per hour. At this speed an output of a little under 0·1 h.p. will be common for level riding. At higher outputs of 0·2 to 0·3 and onwards h.p., for fast efficient cycling the bicycle gearing is such that pedalling speeds of up to 100 revs/min are entailed.

An additional factor in the form of saddle position is well recognized as important to efficient pedalling (see Vaughan Thomas 1967, Dickenson 1927, and Report of the Bicycle Production and Technical Institute, Japan, 1968).

It appears that very large metabolic heat expenditures are involved in fast cycling in unpaced time trials of distances of 100 miles and the 12 and 24 hour rides. In round figures 5,000, 10,000 and 20,000 calories respectively can be involved in achieving the commonly recorded speeds of 25 and 20 miles per hour. Even riders in 'Tourist Trials' covering the standard 100 miles in 8 hours use about 4000 calories. All these strenuous activities bring about a craving for food not satisfied until a day or two's heavy eating has been indulged in by the participants.

It is interesting to note that the relatively high power outputs given as estimates in Table 1 are supported by the fact that engines of about 1⅓ h.p. are fitted to 'mopeds' which have maximum speeds on the level of about 30 miles per hour. Such machines are lightweight motor-cycles (75 lb) and with an average-sized rider will present a frontal area not dissimilar to the 5·5 sq ft assumed for calculating the data in Table 1 for a touring cyclist: these data, when extrapolated to 30 miles per hour, predict a power requirement of 1·05 h.p.

4.2. *Time of Swing of a Leg for Walking and Pedalling*

Grieve and Gear (1966) found that in adult walking actions the time of swing of a leg varied between extreme limits of 0·28 to 0·81 seconds, with a reasonable average of about 0·5 seconds. A pedaller at 50 rev/min swings a leg twice per

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revolution of pedals and cranks, thus has a time of swing of 0·6 seconds. This is remarkably similar to that of the walking action and may be an influence leading to subjects riding ergometer machines accepting 50 revs/min as a ' comfortable ' rate. The power output necessary for a common walking speed of about 3 miles per hour (Dean 1966 and Bekker 1952) is just under 0·1 h.p., as is the power associated with the pedalling rate quoted.

4.3. *Heat Transfer*

The higher power outputs which can be produced by trained cyclists when exercising on ergometers for prolonged periods do not appear to match up to the estimated power outputs based on actual cycling performances. Wilkie, University College London, has listed published information concerning the powers and duration of tolerance associated with trained cyclists exercising on ergometers (see Nonweiler 1956). The highest power output tolerable for a long period, 270 minutes, is recorded as 0·19 h.p. This is much lower than the 0·35 h.p. estimated as necessary to propel a cyclist at 23 miles per hour, a speed which is commonly associated nowadays with cyclists riding in 100 mile distance time trials. Recorded performances on ergometers show that at 0·35 h.p. a brief period only of about 30 minutes can be tolerated.

A feasible explanation for the above discrepancies in power outputs appears to lie in the fact that the riding cyclist is working under cooler and therefore more comfortable conditions than is the pedaller on an ergometer. By virtue of his motion at 23 miles per hour the cyclist is subjected to a forced air flow at some 30-fold rate that caused by free convection effects around the stationary pedaller.

McAdams (1942) gives information concerning the effect of rate of air flow on the magnitude of heat and mass transfer (i.e. evaporation of water to air) film coefficients for the cooling of hot cylindrical objects (the human body for these purposes can be considered as an assembly of cylindrical shapes). It is predictable that an increase of bulk air flow rate, crosswise over cylinders, of 30-fold should increase the magnitude of the film coefficients $(30)^{0.6}$ or some 8-fold. The overall increase in cooling effect produced as a consequence should be adequate to deal with the extra heat associated with a power output of 0·35 as compared with 0·19 h.p. and leave the actual body temperature unchanged and therefore still tolerable to the cyclist.

The proportion of the total output of mechanical power from a cyclist which is used to overcome wind resistance when riding on the level is very high. Detailed calculations show that at 10 miles per hour a crouched racing cyclist, for example, will expend about 50% of his effort in overcoming wind resistance and at 23 miles per hour about 80%. After taking into account the resistance caused by the bicycle itself the air power impact on the cyclist's body can be estimated to be 0·022 and 0·22 respectively horsepower which are measures of the self-created cooling effects caused by his motion through the air.

On a évalué la dépense énergétique nécessaire pour propulser un cycliste assis sur son bicycle et se déplaçant à des vitesses allant jusqu'à la vitesse maximum possible, en exercice prolongé, à cadence libre et sur route plate. Deux méthodes de calcul sont présentées, l'une est basée sur la consommation d'oxygène chez des cyclistes se déplaçant effectivement sur une route et que l'on trouve dans la littérature, l'autre est basée sur des données enregistrées concernant la résistance à la traction que présentent des cyclistes en mouvement.

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Der Energieaufwand wird geschätzt, der zum Antrieb aufgesessener Radfahrer bei Geschwindigkeiten bis zur maximalen, für längere Fahrperioden ohne Schrittmacher auf ebener Strasse möglichen, nötig ist. Zwei Berechnungsmethoden werden vorgestellt: die eine beruht auf dem in der Literatur veröffentlichten Sauerstoffverbrauch von Radfahrern, die tatsächlich auf der Strasse fuhren, die andere stützt sich auf vorliegende Daten über den Zugwiderstand bei der Bewegung aufgesessener Radfahrer.

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